

Species distribution modelling repository for KwaZulu-Natal

Work flow overview.

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config:
  layout: elk
  elk:
    nodePlacement.strategy: SIMPLE
    padding: 25
    spacing.nodeNode: 50          # ↑ increase node spacing
    spacing.nodeNodeBetweenLayers: 50
    algorithm: layered
  themeVariables:
    primaryTextColor: "#322c2c"    # arrow labels
    textColor: "#312709"           # node text
    labelTextColor: "#151319"      # node labels
    lineColor: "#e69d1e"           # arrow color
    edgeLabelBackground: "#e2c888bf" # link bg color
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flowchart TD
    SpeciesRecordDB["Species Record<br/>Database"] -->|Pull species<br/>occurrence data| DataInput
    DataInput --> DecisionPoint["Decision: <br/>Analysis Type"]

    DecisionPoint -->|Simple intersection| PointIntersection["Point Intersection<br/>with Planning Units"]
    DecisionPoint -->|Environmental<br/>overlay| EnvOverlay["Environmental<br/>Overlay Analysis"]
    DecisionPoint -->|Full SDM<br/>modelling| SDMAAnalysis["Species Distribution<br/>Model Analysis"]

    SpeciesRecordDB -->|Query species<br/>traits & groups| SpeciesTraits["Species Traits<br/>Database"]
    SpeciesTraits --> EnvDataQuery["Query Environmental<br/>Data by Species<br/>Traits/Groups"]

    EnvDataDB["Environmental Data<br/>Database"] -->|Fetch layers:<br/>climate, soil,<br/>vegetation| EnvDataQuery

    EnvDataQuery --> SDMAAnalysis
    EnvDataQuery --> EnvOverlay

    PointIntersection --> Output["Species<br/>Distribution<br/>Map"]
    EnvOverlay --> Output
    SDMAAnalysis --> Output

    Output --> PlanningUnits["Planning Units<br/>Layer"]
```

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classDef database fill:#618571,stroke:#9da19f
classDef process fill:#dda381,stroke:#909592
classDef decision fill:#fde68a,stroke:#fb923c
classDef output fill:#618571,stroke:#9da19f
classDef envdecision fill:#fde68a,stroke:#fb923c

class SpeciesRecordDB,EnvDataDB database
class DataInput,SpeciesTraits,EnvDataQuery,PointIntersection,EnvOverlay,SDMANalysis process
class DecisionPoint decision
class envDecisionPoint envdecision
class Output,PlanningUnits output
```

option a: